

Deep Learning

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Logistic Regression

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We choose cross entropy distance as our loss function:

$$\frac{1}{n} \sum_{l=1}^n (-y_l \log f(x_l) - (1 - y_l) \log (1 - f(x_l))).$$

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- 3 Why don't we use L^2 distance in this case?
- 4 Can we do logistic regression when number of classes is greater than 2?

L1 and L2 Regularizations

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In linear regression instead of L^2 loss we use one from this two:

$$\frac{1}{n} \sum_{l=1}^n (f(x_l) - y_l)^2 + \frac{\lambda}{k} \sum_{l=1}^k |w_l|,$$

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1 Softmax Classifier

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- 3 What to do in the case of multi-label classification?